

TTBO-FL: Joint Training, Trajectory, and Beamforming Optimization for Energy-Efficient Federated Learning in UAV Swarm

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Abstract—Unmanned aerial vehicles (UAVs) can use the collected data to perform machine-learning tasks and enhance their intelligence level. The distributed framework of federated learning (FL) is suitable for resource-constrained UAV swarms. However, the dynamic channel conditions caused by the mobility of the UAVs impact the wireless FL performance. Additionally, data insufficiency and heterogeneity also affect the performance of the trained models. Thus, in this article, we propose a joint training, trajectory, and beamforming optimization for an energy-efficient FL scheme that leverages the mobility of UAVs to enhance the model training efficiency, namely, TTBO-FL. We adopt a two-level FL for the UAV network, where a high-level UAV (H-UAV) is for model aggregation and a set of low-level UAVs (L-UAVs) is for model training. Considering the dynamic channel conditions, we design a 3-D uniform linear array (3-D ULA) and implement 3-D analog beamforming to increase communication efficiency among UAVs. Moreover, we introduce transfer learning techniques and regularization constraints to mitigate the problem of data insufficiency and heterogeneity. Then, we formulate an optimization problem for UAV trajectory planning, local epoch adjustment, and beamforming and adopt a soft actor-critic (SAC)-based algorithm to solve it. The simulation results show that, compared to the baseline schemes, the proposed schemes achieve higher model accuracy and lower energy consumption for the UAV swarm.

Index Terms—Beamforming, federated learning (FL), trajectory planning, unmanned aerial vehicle (UAV) swarm.

I. INTRODUCTION

DUE to the mobility, flexibility, and favorable line-of-sight (LoS) link [1], [2], [3], unmanned aerial vehicles (UAVs) have been widely used in various fields, such as forest fire detection [4], air quality index (AQI) monitoring [5], object tracking [6], and area coverage [7]. The deployment of machine learning (ML) models within UAV swarms further enhances their level of intelligence, with these models being

trained on data collected by UAVs to enhance their performance [8], [9]. Traditional ML solutions require UAVs to transmit raw data to a central entity for centralized processing [10]. However, for resource-constrained UAVs, transmitting raw data consumes a significant amount of energy and introduces long delays [11], impacting model convergence speed and task execution. In addition, the transmission of raw data can raise privacy concerns. Furthermore, when UAVs perform tasks in remote areas, infrastructure such as base stations may be inaccessible.

As a distributed learning framework, federated learning (FL) [12] proposes transmitting model parameters instead of raw data, learning a shared global model from distributed mobile devices [13], thus preserving privacy while reducing communication costs. The framework is highly suitable for UAVs with limitations in computation, bandwidth, and energy, which has been widely applied in UAV swarms. However, deploying FL in UAV swarms faces some challenges, such as dynamic channel conditions, statistical heterogeneity of data, and resource constraints. The mobility of UAVs causes channel variations that can impact the exchange of model parameters, thereby affecting the convergence and model performance of FL. Additionally, the limited range and diversity of flying areas result in UAV swarms collecting data that is insufficient, imbalanced, and non-independent and identically distributed (non-IID), which adversely impacts FL performance. Finally, UAV swarms are constrained by limited bandwidth, computing capabilities, and energy resources. Consequently, optimizing the overall design of an energy-efficient FL framework in UAV swarms is essential to enhance its performance.

Leveraging the mobility of UAV swarms, we propose an energy-efficient FL framework based on trajectory planning for UAV swarms. Specifically, we consider a two-level UAV communication network comprising a high-level UAV (H-UAV) and a group of low-level UAVs (L-UAVs), where the H-UAV is responsible for FL model aggregation and swarm control while L-UAVs conduct local model training in FL and execute intelligent tasks. We enhance communication efficiency between UAVs through beamforming design and then improve the model performance of FL through transfer learning techniques and regularization constraints. Finally, we propose an FL optimization scheme based on trajectory planning, achieving high energy efficiency in FL. The main contributions of this article can be summarized as follows.

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- 1) We propose a two-level UAV communication network for FL, utilizing transfer learning to deploy pretrained models on L-UAVs and regularization constraints to mitigate the effects of data scarcity and heterogeneity.
- 2) We develop a scheme that jointly considers FL training strategies, trajectory, and beamforming for H-UAVs, namely, TTBO-FL. Specifically, we design a 3-D uniform linear array (3-D ULA) and implement 3-D analog beamforming to enhance communication efficiency among UAVs. Moreover, local epoch adjustment is designed to balance training delays and model performance.
- 3) We formulate an optimization problem integrating trajectory planning, beamforming for H-UAVs, and local epoch adjustment and then propose a soft actor-critic (SAC) algorithm to solve the above problem.
- 4) Simulation results show that our proposed scheme and algorithm effectively improve the model accuracy while minimizing the energy consumption of H-UAV for FL deployment in UAV networks compared to the baseline methods.

The remaining part of this article is organized as follows. In Section II, we review related work and methods. Section III introduces the system model and problem description. In Section IV, we formulate the problem as an MDP and propose our algorithm. In Section V, we present simulation results and performance analysis, followed by conclusions in Section VI.

II. RELATED WORKS

FL in the domain of UAV swarms has recently been widely researched. In this section, we discuss the related works concerning FL deployment in UAV swarms and trajectory planning in FL-enabled UAVs.

A. FL in UAV Swarms

Zeng et al. [14] proposed an FL framework for UAV swarms to minimize the convergence rounds by jointly optimizing power allocation and UAV scheduling. Shen et al. [15] proposed a two-level FL framework for UAV swarm, which aims to minimize communication and computation energy consumption of all training UAVs by jointly optimizing local convergence threshold, computation resource, and bandwidth allocation. Yang et al. [16] proposed a UAV-assisted FL framework, where UAVs act as servers to assist devices for FL training, by solving a joint device selection, UAV placement, and resource allocation problem to reduce FL convergence time and improve model accuracy. Liu et al. [5] proposed an FL-based aerial-ground air quality sensing framework by deploying FL in the UAV swarm and combining it with the ground wireless sensor network to achieve low energy consumption and accurate AQI inference. Song et al. [17] introduced a UAV-assisted FL framework based on nonorthogonal multiple access (NOMA) by optimizing the uplink NOMA transmission duration, downlink broadcast duration, and computation rates of UAVs to minimize the FL training delay while improving model accuracy. Liu et al. [18] proposed an FL framework for UAV swarms and investigated a

resource allocation problem aimed at jointly optimizing computation resource allocation and UAV association to reduce the training delay and energy consumption in FL. Yahya and Maghsudi [19] introduced a UAV-enabled FL network and solved a joint optimization problem concerning UAV positioning, channel allocation, and power control to minimize the convergence time of FL.

By introducing intermediate aggregation layers—typically at UAV clusters or edge servers, hierarchical FL (HFL) plays a critical role in enhancing the efficiency, scalability, and robustness of FL systems within UAV networks. Wang et al. [20] proposed a two-tier HFL scheme, where the UAVs act as relays to assist the edge server and local clients in model training, improving the training and communication efficiency. Tur-sunboev et al. [21] proposed an HFL-assisted UAV network, which exploits the edge servers as intermediate aggregators with shared data, reducing the training latency and overcoming the problem of time-varying data heterogeneity. Li et al. [22] proposed an HFL framework to enhance the robustness and efficiency by implementing both intragroup and global model aggregation. Zhagypar et al. [23] proposed an unbiased HFL framework for UAV-assisted networks that relies on UAVs for local aggregation to mitigate the impact of unreliable channels between edge devices and servers. Li et al. [24] proposed a hierarchical deep FL, which uses capacity values as a criterion to select the leader UAV to become a top-level server for model aggregation to maximize traffic offloading and minimize UAV energy consumption simultaneously.

B. Trajectory Planning in FL-Enabled UAVs

Trajectory planning for UAVs has always been a hot research topic in this field. Zhong et al. [25] proposed a UAV-assisted hierarchical aggregation over-the-air FL (OA-FL) system that improves the FL learning performance by adjusting the UAV trajectory and global aggregation coefficients. Mestoukirdi et al. [26] employed UAVs as mobile orchestrators to coordinate the transmissions and the training schedule among the devices and accelerated the training process of all FL tasks by optimizing the UAV trajectory and device selection. Hou et al. [27] introduced hierarchical nested personalized FL (HN-PFL). At the micro level, UAVs within swarms were orchestrated in a distributed manner to optimize energy consumption and ML model performance. At the macro level, the swarm trajectories and the training duration of HN-PFL were optimized to enhance the ML performance. To further improve data security, Wang et al. [28] proposed a UAV-enabled covert FL architecture, balancing security and training costs by jointly optimizing the trajectory of the UAV, computation and communication resource allocation, as well as local model accuracy requirements.

However, the aforementioned works did not account for the dynamic variability of UAV trajectories driven by tasks, as well as the non-IID characteristics of data on UAVs and the impact of insufficient data on FL model training. Furthermore, they also failed to consider the flight energy consumption of the UAV, which constitutes a significant portion of the energy consumption during the UAV's movement.

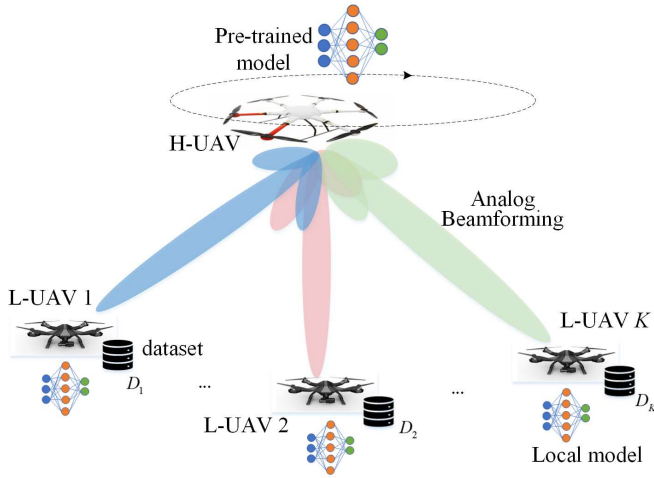


Fig. 1. FL based on trajectory planning in the UAV swarm.

III. SYSTEM MODEL AND PROBLEM FORMULATION

Fig. 1 illustrates the two-level UAV network with FL, including the H-UAV and a group of L-UAVs. In this scenario, L-UAVs execute some intelligent tasks and generate task-oriented trajectories. The UAV swarm can be deployed with pretrained neural network models before executing intelligent tasks. Additionally, they usually collect some data, such as images of targets or trajectory data, during the execution of tasks, and could participate in collaborative training of neural network models to increase the performance of models. H-UAV is responsible for model parameter aggregation and control of the UAV swarm (e.g., swarm flight heading).

Let us represent the L-UAVs set as $\mathcal{K} = \{1, \dots, k, \dots, K\}$. We consider a finite execution time of FL, divided into N time slots, denoted as $n \in \mathcal{N} = \{1, \dots, n, \dots, N\}$. In a time slot n whose length is denoted as t^n , since t^n is appropriate, the H-UAV has enough time to make a better decision, and the channel power gain between UAVs is not changed largely [16]. In the n th time slot, let us denote the positions of the H-UAV and L-UAV k as $\mathbf{q}_H^n = (X_H^n, Y_H^n, H)$ and $\mathbf{q}_k^n = (x_k^n, y_k^n, h)$, respectively. The distance between the H-UAV and L-UAV k is represented as $d_k^n = \|\mathbf{q}_H^n - \mathbf{q}_k^n\|_2$.

A. Federated Learning Model

In the FL process, we denote $\mathbf{w}^n$ as the global model parameters for the n th communication round and $\mathbf{w}_k^n$ as the local model parameters for the L-UAV k in time slot n . We represent $\mathcal{D}_k = \{\mathbf{x}_{k1}, \mathbf{x}_{k2}, \dots, \mathbf{x}_{kD_k}\}$ as the local train dataset for the L-UAV k , where $D_k = |\mathcal{D}_k|$ is the data size, and each data sample $\mathbf{x}_{kd}$ corresponds to its sample label y_{kd} . For the FL task of UAV swarm, we aim to optimize the global model by minimizing the loss function as follows:

$$\min_{\mathbf{w}} F(\mathbf{w}) \triangleq \sum_{k=1}^K \frac{D_k}{D} F_k(\mathbf{w}) = \frac{1}{D} \sum_{k=1}^K \sum_{d=1}^{D_k} f(\mathbf{w}; \mathbf{x}_{kd}, y_{kd}) \quad (1)$$

where $f(\mathbf{w}; \mathbf{x}_{kd}, y_{kd})$ is the loss of sample data $\{\mathbf{x}_{kd}, y_{kd}\}$ with model parameter $\mathbf{w}$. The specific process of a communication round in FL is outlined as follows.

- 1) *Global Model Broadcasting*: The H-UAV distributes the current global model parameters to the L-UAVs through broadcasting in time slot n . Simultaneously, it broadcasts relevant control information such as the local epoch. In addition, to improve training convergence under data insufficiency and heterogeneity, transfer learning is employed. In the initial round or when new L-UAVs join the network, their local models are initialized from a pretrained model $\mathbf{w}^{\text{pre}}$ rather than from zero. The overall step is formulated as

$$\mathbf{w}_k^n \leftarrow \begin{cases} \mathbf{w}^{\text{pre}}, & \text{if } n = 0 \text{ or UAV } k \text{ is new} \\ \mathbf{w}^n, & \text{otherwise.} \end{cases} \quad (2)$$

This transfer-based initialization reduces the optimization gap between global and local models, improves gradient alignment, and accelerates convergence.

- 2) *Local Model Training*: In this step, the L-UAVs that have received the global model parameters perform local training for several epochs based on their local datasets. Typically, this involves updating the model parameters using mini-batch stochastic gradient descent (SGD). Due to the non-IID nature of data across different UAVs, overfitting and data drift may occur during local training [29], which degrades global model convergence. To address this, we incorporate regularization mechanisms in our local training process. Inspired by [30], we apply feature decorrelation to mitigate the adverse effects of dimensional collapse caused by data heterogeneity in FL. This is achieved by adding the following regularization term to the loss function during local training:

$$\min_{\mathbf{w}} F_k(\mathbf{w}; \mathbf{X}, \mathbf{y}) = l(\mathbf{w}; \mathbf{X}, \mathbf{y}) + \beta \frac{\|R_{\text{corr}}\|_F^2}{d_{\text{rep}}^2} \quad (3)$$

where $\mathbf{X}$ and $\mathbf{y}$ represent the input matrix of mini-batch samples and the target label matrix, respectively; $\mathbf{w}$ denotes the model parameters; $l(\cdot)$ represents the original loss function, which typically indicates the cross-entropy loss function in classification tasks; β denotes the coefficient of the regularization term; d_r represents the dimension of the feature matrix extracted by the neural network from the samples; R_{corr} represents the correlation matrix of the feature matrix for mini-batch samples; and $\|\cdot\|_F$ represents the Frobenius norm. This regularization encourages diverse feature representations and mitigates overfitting on biased or small datasets, helps stabilize training under heterogeneous data conditions, and accelerates convergence by reducing the impact of data drift and improving feature generalization. After several epochs, the update of local model for L-UAV k is expressed as $\mathbf{w}_k^{n+1} = \mathbf{w}_k^n - \lambda \nabla F_k(\mathbf{w}_k^n)$, where λ is expressed as the learning rate.

- 3) *Model Aggregation*: All L-UAVs upload their local model parameters via the uplink to the H-UAV. Upon receiving these local model parameters, the H-UAV proceeds with model aggregation. Typically, weighted aggregation is used to better represent the contributions of L-UAVs to the model parameters. This includes methods like weighted gradient, weighted loss, and weighted

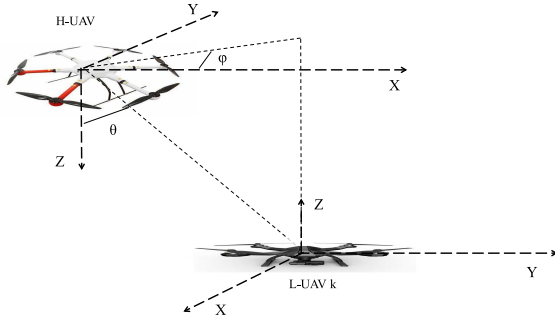


Fig. 2. Schematic illustrating the pitch and yaw angles.

data size, where weighted gradient and weighted loss aggregation require computing and uploading the gradient or loss information from L-UAVs at the current communication round. To reduce computational complexity, we perform weighted data size aggregation for the local model parameters of all L-UAVs, where the local data size remains unchanged during the FL process, denoted as

$$\mathbf{w}^{n+1} = \frac{\sum_{k \in \mathcal{K}} D_k \mathbf{w}_k^{n+1}}{\sum_{k \in \mathcal{K}} D_k}. \quad (4)$$

B. Analog Beamforming

To improve energy efficiency in UAV communication networks [31], we assume that all UAVs are equipped with directional antennas, leveraging analog beamforming to enhance channel quality while minimizing communication cost, power consumption, and dependence on real-time channel state prediction [32]. In addition, the adopted directional antenna is assumed to be operated with a realistic 3-D ULA for power concentration.

In 3-D ULA, isotropic antenna elements are evenly placed in 3-D space with equal spacing $\lambda_0/2$, where λ_0 represents the wavelength of the RF signal's center carrier. The antenna elements of each axis are represented as A_x, A_y , and A_z and the total number of elements in the antenna array is $A_t = A_x \times A_y \times A_z$. The array factor of the 3-D ULA can be formulated as [32]

$$F_{\text{array}} = \frac{\sin(A_x \frac{\pi}{2} \phi_x)}{A_x \sin(\frac{\pi}{2} \phi_x)} \frac{\sin(A_y \frac{\pi}{2} \phi_y)}{A_y \sin(\frac{\pi}{2} \phi_y)} \frac{\sin(A_z \frac{\pi}{2} \phi_z)}{A_z \sin(\frac{\pi}{2} \phi_z)} \quad (5)$$

where the auxiliary variables are

$$\begin{aligned} \phi_x &= \sin \theta \cos \varphi + \alpha_x \\ \phi_y &= \sin \theta \sin \varphi + \alpha_y \\ \phi_z &= \cos \theta + \alpha_z \end{aligned} \quad (6)$$

where θ and φ are, respectively, the elevation and azimuth angles between the H-UAV and L-UAVs, as illustrated in Fig. 2.

For analog beamforming, the uniform array in the 3-D ULA requires a linearly progressive phase among the antenna elements [33]. $\alpha_x, \alpha_y, \alpha_z \in [-1, 1]$ are the factors related to phase shift among antenna elements in analog beamforming, which can be changed by phase shifters to control the main beam of the antenna array toward a particular direction [34]. In

addition, according to the relationship among elevation angle, azimuth angle, and rectangular coordinates, the auxiliary variables can also be expressed as

$$\begin{aligned} \phi_x &= \frac{X_H^n - x_k^n}{d_k^n} + \alpha_x \\ \phi_y &= \frac{Y_H^n - y_k^n}{d_k^n} + \alpha_y \\ \phi_z &= \frac{H - h}{d_k^n} + \alpha_z. \end{aligned} \quad (7)$$

The antenna gain is the function of the array factor, which is expressed as

$$\begin{aligned} G &= A_t \underbrace{\left(\frac{\sin(A_x \phi_x^*)}{A_x \sin(\phi_x^*)} \right)^2}_{g_x} \underbrace{\left(\frac{\sin(A_y \phi_y^*)}{A_y \sin(\phi_y^*)} \right)^2}_{g_y} \underbrace{\left(\frac{\sin(A_z \phi_z^*)}{A_z \sin(\phi_z^*)} \right)^2}_{g_z} \\ &= A_t g_x g_y g_z \end{aligned} \quad (8)$$

where $\phi_i^* = \pi/2 \phi_i, i \in \{x, y, z\}$. When CSI is imperfect, the actual position or channel deviation leads to inaccurate estimation of ϕ_i , i.e., $\phi_i = \phi_i^{\text{true}} + \epsilon_i$, introducing steering misalignment. This misalignment affects the main-lobe direction and gain, possibly leading to sidelobe beamforming or partial signal loss. To quantify this, we adopt an antenna gain approximation [14]

$$\widehat{g}_i(\phi_i) = \begin{cases} g_i(\phi_i), & |\phi_i| < \frac{-2\delta}{A_i} \\ G_{\min}, & |\phi_i| \geq \frac{-2\delta}{A_i} \end{cases} \quad (9)$$

where $\delta \in (0, 1)$ is the modification factor, with a larger δ implying a better approximation of the main lobe. $G_{\min}$ is set to the gain of the first sidelobe. By taking the derivative of $g_i(\phi_i)$ and getting the gain of the first sidelobe, $G_{\min} = 1/A_i^2$ is obtained. Additionally, the analog beamforming scheme in time slot n is denoted as $\alpha^n = (\alpha_x^n, \alpha_y^n, \alpha_z^n)$. This approximation captures the performance drop when the true direction falls outside the main lobe. A larger δ implies better robustness, but increasing δ also broadens the main lobe, sacrificing beam precision. Thus, while imperfect CSI may degrade antenna gain to a certain extent (from G to $G_{\min}$), the overall FL performance degradation is constrained.

C. Communication Model

The communication related to UAVs mainly involves three types: UAV-to-ground base station communications, UAV-to-ground terminal communications, and UAV-to-UAV communications. Communication between UAVs is referred to as air-to-air (A2A) channels, typically modeled as a simple free-space path loss model [35], [36]. To facilitate the analysis and evaluation of energy consumption for UAV swarm in the FL process, we model the A2A channel as a free-space propagation channel model, where the effects of shadow fading and small-scale fading are neglected [37]. The channel power gain is simplified as

$$h_0(d_k) = \left(\frac{\lambda_0}{4\pi d_k} \right)^2 = \rho_0 d_k^{-2} \quad (10)$$

where λ_0 is the carrier wavelength and $\rho_0 = (\lambda_0/4\pi)^2$ is the channel power gain at unit distance. Therefore, based on the aforementioned channel model, the channel power gain is determined by the distance d_k between the H-UAV and the L-UAV k . It is assumed that L-UAVs use orthogonal frequency division multiple access technology for local model uploading [38], while the H-UAV transmits the global model via broadcasting. Therefore, the channel capacities for the uplink and downlink can be, respectively, expressed as

$$R_k^{\text{up}} = B_{\text{sub}} \sum_{m=1}^M \chi_{k,m} \log_2 \left(1 + \frac{p_k \rho_0 G_a}{d_k^2 \sigma^2} \right) \quad (11)$$

$$R_k^{\text{down}} = B \log_2 \left(1 + \frac{p_H \rho_0 G_a}{d_k^2 \sigma^2} \right) \quad (12)$$

where G_a is the antenna gain, including both the transmit and receive antenna gains. Since L-UAVs only communicate with the H-UAV for model parameters exchange, it can be assumed that the transmit antenna gain for the uplink and the receive antenna gain for the downlink are both main-lobe gains. B_{sub} is the subchannel bandwidth; M is the number of subchannels; B is the broadcast bandwidth for the H-UAV; σ^2 is the noise power; and p_H and p_k are the transmit powers of the H-UAV and the L-UAV k , respectively. In addition, if the m th subchannel is allocated to L-UAV k , then the indicator factor $\chi_{k,m} = 1$; otherwise, $\chi_{k,m} = 0$. Therefore, the uplink and downlink transmission times are expressed as $t_k^{\text{up}} = S_1/R_k^{\text{up}}$ and $t_k^{\text{down}} = S_2/R_k^{\text{down}}$, where S_1 and S_2 represent the transmitted data size in the uplink and downlink, respectively.

D. Computation Model

Whether local model training or global model aggregation, it heavily relies on the central processing unit (CPU) as the primary engine of computation. CPU performance is controlled by the CPU cycle, also known as the CPU clock speed. During the local model training, L-UAV k updates the local model using SGD. Define C_k as the number of CPU cycles required to train the model on one sample data for L-UAV k and C_H as the number of CPU cycles required to process per data bit for H-UAV. Let f_H and f_k represent the computation capability of H-UAV and L-UAV k , respectively. Therefore, the computation latency of H-UAV and L-UAV k can be, respectively, defined as $t_H^{\text{comp}} = K S_1 C_H / f_H$ and $t_k^{\text{comp}} = I D_k C_k / f_k$.

E. Flying Model

Given the flexibility of rotary-wing UAVs, we deploy multi-rotor UAVs in the scenario and formulate the flight and energy consumption models. Let $v^n = \|\mathbf{v}^n\|$ be the scalar velocity and θ^n be the angle of the vector velocity $\mathbf{v}^n$ with respect to the x-axis for the H-UAV. Therefore, the coordinate change of the H-UAV is expressed as

$$\Delta X_H^n = v^n t^n \cos \theta^n \quad (13)$$

$$\Delta Y_H^n = v^n t^n \sin \theta^n. \quad (14)$$

The power consumption of a UAV during flight is generally related to its velocity and acceleration. However, the energy consumption caused by the acceleration and deceleration of

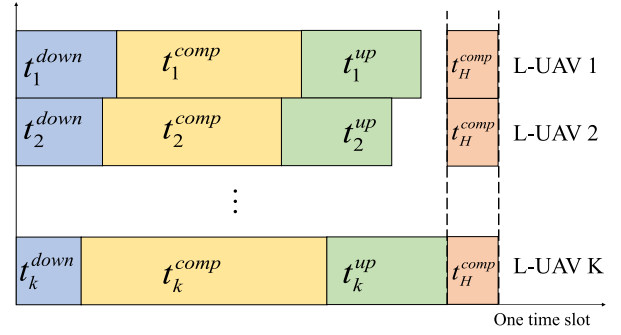


Fig. 3. Time slot diagram of a communication round.

UAVs is typically minimal because the duration of acceleration and deceleration usually accounts for only a small fraction of the total flight time of UAVs. Therefore, in most cases, the power consumption of a rotary-wing UAV can be represented as the univariate function of its flight velocity v , which is specifically modeled as [39]

$$P(v) = P_0 \left(1 + \frac{3v^2}{U_{\text{tip}}^2} \right) + P_1 \sqrt{\sqrt{1 + \frac{v^4}{4v_0^4}} - \frac{v^2}{2v_0^2} + \frac{1}{2}\zeta s \rho A v^3} \quad (15)$$

where P_0 and P_1 represent the blade profile power and induced power in hovering status, respectively; U_{tip} and v_0 represent the tip speed of the rotor blade and the mean rotor-induced velocity when hovering, respectively; ζ and s denote, respectively, the fuselage drag ratio and rotor solidity; ρ is the air density; and A represents the rotor disk area.

F. Problem Formulation

In the proposed framework, we investigated joint optimization of the trajectory of the H-UAV, analog beamforming, and local epoch adjustment, with the aim of minimizing the weighted sum of energy consumption for the H-UAV and loss of the global model over a certain number of communication rounds N . According to the FL process, the length of the n th time slot can be calculated as

$$t^n = \max_{k \in K} \left(t_k^{\text{down},n} + t_k^{\text{comp},n} + t_k^{\text{up},n} \right) + t_H^{\text{comp},n}. \quad (16)$$

In a communication round, each L-UAV has its own download and upload communication latency, as well as local computational latency, while the H-UAV collects the parameters uploaded by all L-UAVs and then performs model aggregation, i.e., the H-UAV performs model aggregation when it receives parameters from the last L-UAV. The slot diagram is illustrated in Fig. 3.

The total energy consumption of UAVs mainly consists of two parts. The first one is energy consumption related to communication and computation. The other is the propulsion energy consumption, which is necessary to ensure that UAVs remain aloft and to support their maneuverability [39]. In practice, energy consumption related to communication and computation is typically much smaller than propulsion energy consumption, such as a few watts compared to several hundred watts [40]. Therefore, we disregard communication and

computation energy consumption. The energy consumption of the H-UAV is calculated as $E_n = t^n P(v^n)$. Furthermore, the performance of the global model is defined as the final loss change for the global model $\Delta F(\mathbf{w}) = F^N(\mathbf{w}) - F^0(\mathbf{w})$, with $F^0(\mathbf{w})$ representing the initial loss. We consider κ as the weight of energy consumption and global model loss change.

Thus, the problem of minimizing system cost can be expressed as

$$\begin{aligned} \mathbf{P0} : & \min_{\mathbf{v}, \theta, \mathbf{I}, \alpha} \sum_{n=1}^N \kappa t^n P(v^n) + \Delta F(\mathbf{w}) \\ \text{C1} : & 0 \leq \theta^n \leq 2\pi \quad \forall n \in \mathcal{N} \\ \text{C2} : & v_{\min}^T \leq v^n \leq v_{\max}^T \quad \forall n \in \mathcal{N} \\ \text{C3} : & \Delta X_H^n = v^n t^n \cos \theta^n \quad \forall n \in \mathcal{N} \\ \text{C4} : & \Delta Y_H^n = v^n t^n \sin \theta^n \quad \forall n \in \mathcal{N} \\ \text{C5} : & \alpha_x^n, \alpha_y^n, \alpha_z^n \in [-1, 1] \quad \forall n \in \mathcal{N} \\ \text{C6} : & I_{\min} \leq I^n \leq I_{\max} \quad \forall n \in \mathcal{N}. \end{aligned} \quad (17)$$

In problem **P0**, C1 and C2 represent the flight direction and speed range of the H-UAV, respectively. C3 and C4 represent the constraints on the coordinate changes of the H-UAV. C5 represents the range of analog beamforming variables. C6 represents the range of local epoch I .

IV. PROPOSED SAC-BASED SOLUTION

In this section, we introduce the proposed SAC-based solution, which includes reformulating the optimization problem **P0** of dynamic strategies as a Markov decision process (MDP). In addition, we present the design of the proposed SAC algorithm. Leveraging the robust pattern-matching capabilities of SAC, we address our sequential decision-making problems.

A. Modeling of Reinforcement Learning Environment

To efficiently deploy FL for model training in UAV swarms, we need to jointly optimize the trajectory of the H-UAV, analog beamforming, and local epoch. Given the highly dynamic nature of UAV trajectories and channel conditions in the FL process, we reformulate the optimization problem **P0** as an MDP, which includes state space $\mathcal{S}$, action space $\mathcal{A}$, and reward function r to employ the SAC algorithm. Agents select and execute actions based ON current environmental states, which influence the transition and updating of the environment state, determining the reward feedback received by the agents.

1) *State Space*: The quality of state information is fundamental for agents to devise strategies and consider long-term benefits. The design of the state space significantly influences the convergence speed and final performance of the RL algorithm. We construct a 3-D coordinate point for each L-UAV, denoted as $\mathbf{q}_k^n$. To expedite convergence, the state space also includes the 3-D coordinates of the H-UAV, denoted as $\mathbf{q}_H^n$, as well as the pitch and yaw angle information of L-UAV k relative to the H-UAV, denoted as $\Phi_k^n = (\sin \theta_k^n \cos \varphi_k^n, \sin \theta_k^n \sin \varphi_k^n, \cos \theta_k^n)$. Considering the dynamic nature of the environment, in the n th iteration of FL, the next coordinate point of L-UAV k is associated with its movement characteristics in a specific scenario, following

constraints similar to those of the H-UAV. Meanwhile, $\mathbf{q}_H^n$ is related to the trajectory planning of the H-UAV, and Φ_k^n is associated with the coordinates of both the H-UAV and L-UAV k . Although incorporating more state information could accelerate convergence, considering practical scenarios where additional state information, such as UAV battery levels and current model performance metrics, may be challenging to obtain, we refrain from adding extra state information. Hence, the state space can be represented as follows:

$$\mathcal{S}_n = \{\{\mathbf{q}_k^n, \alpha_k^n\}_{k \in \mathcal{K}}, \mathbf{q}_H^n\}_{n \in \mathcal{N}}. \quad (18)$$

2) *Action Space*: When designing the action space, it is necessary to consider feasible actions to achieve the objectives and avoid action dead zones. In order to reduce the complexity of the action space and take into account the parameters of the optimization problem, the action space involves four optimization variables, including the flight velocity v^n and flight direction θ^n that control the trajectory of the H-UAV, the local epoch I^n , and the variables α^n related to analog beamforming. Therefore, the action space is defined as

$$\mathcal{A}_n = \{v^n, \theta^n, I^n, \alpha^n\}_{n \in \mathcal{N}}. \quad (19)$$

3) *Reward*: In deep reinforcement learning (DRL) algorithms such as SAC, the rewards guide the neural network to seek state- and action-value functions, refined based on environmental characteristics, and are used to compute and generate actions. When determining an action a under state s , the agent receives a reward. During the time slot n , the positions of the L-UAVs are random, causing dynamic channel quality. According to our optimization problem, we consider the weighted negative of flight energy consumption for H-UAV and the change of global model loss as the reward, i.e.,

$$R_n = -\kappa E^n + F^n(\mathbf{w}) - F^{n-1}(\mathbf{w}). \quad (20)$$

Algorithm 1 Proposed State Normalization

- 1 Initialize the Normalizer Z . Including the mean $\mathbf{M}_N = \mathbf{0}$, variance $\mathbf{S}_N = \mathbf{0}$ and counter $n_N = 0$.
 - 2 **for** each step **do**
 - 3 Get the states
 - 4 Update the counter $n_N \leftarrow n_N + 1$
 - 5 Update the mean by $\mathbf{M}_s^{\text{new}} = \mathbf{M}_s^{\text{old}} + (\mathbf{s} - \mathbf{M}_s^{\text{old}})/n_s$.
 - 6 Update the variance by $\mathbf{S}_s \leftarrow \mathbf{S}_s + (\mathbf{s} - \mathbf{M}_s^{\text{old}})(\mathbf{s} - \mathbf{M}_s^{\text{new}})$.
 - 7 Return the normalized state $\mathbf{s} \leftarrow (\mathbf{s} - \mathbf{M}_s^{\text{new}})/\sqrt{\mathbf{S}_s/n_s}$
 - 8 **end for**
-

4) *Normalization*: Normalization techniques are widely employed in neural networks to expedite training, such as batch normalization [41] in the field of computer vision and layer normalization [42] in natural language processing. In the design of our SAC algorithm, we perform reward shaping for reward normalization. Specifically, while the model loss remains within the single-digit range without scaling, a scaling factor κ is introduced to bring it to a comparable magnitude as the performance metric loss due to significant energy consumption. This process balances energy consumption and model performance while also normalizing rewards. For continuous action spaces, we incorporate the hyperbolic tangent

(tanh) function at the last layer of the actor network to normalize the output layer. This maps the action space to the range of $[-1, 1]$ while introducing nonlinearity. In feedback from the environment, the normalized actions are restored to the original action space after a linear transformation. Regarding the state space, a simple method involves normalization by multiplying by a fixed weight. However, given the complexity and variability of the state space, it is difficult to find suitable weights. Hence, we adopt an adaptive Gaussian normalization method. The pseudocode for the algorithm flow is depicted in Algorithm 1.

B. SAC-Based Algorithm

We present the proposed SAC-based algorithm to make decisions for each communication round, elucidating how the proposed algorithm seeks equilibrium. As we know, DRL algorithms find extensive application in dynamic scenarios for trajectory planning and resource allocation [43], [44], [45]. Leveraging the robust pattern-matching capabilities of DRL, we address our sequential decision-making problems.

SAC is an off-policy algorithm for maximum entropy DRL [46]. SAC is suitable for handling tasks with continuous action spaces, i.e., $a = \pi(s; \beta^\pi)$, where β^π represents the parameters of the actor network and s and a are the inputs and outputs of the actor network, respectively. It also improves sample efficiency using experience replay while utilizing target networks to reduce overestimation and stabilize the learning process. In the SAC algorithm, the actor network outputs the mean and variance of actions following a Gaussian distribution. Furthermore, it maximizes not only the rewards but also the entropy of the policy, denoted as $\alpha H(\pi(\cdot|s_i))$, where the temperature coefficient α determines the importance of the entropy term relative to the rewards.

The SAC algorithm we designed consists of one actor network and four critic networks, two of which are denoted as Q^1 and Q^2 . In each update, the smaller of the two Q values is chosen to prevent overestimation. The other two critic networks, $Q^{1'}$ and $Q^{2'}$, serve as target networks to stabilize training. Let β^Q denote the parameters of the critic network. The loss function of the critic networks is represented as

$$L^Q = \frac{1}{2c} \sum_{j=1}^2 \sum_{i=1}^c \frac{1}{2} \left(Q_i^j(s_i, a_i; \beta^{Q^j}) - Q_i^t \right)^2 \quad (21)$$

$$Q_i^t = r_i + \gamma \left(\min_{j \in \{1,2\}} Q_i^j(s_{i'}, a_{i'}; \beta^{Q^j}) - \alpha \log(a_{i'}) \right) \quad (22)$$

where $s_{i'}$ represents the next transition state of state s_i and $a_{i'}$ represents the action taken in the next transition state $s_{i'}$. c represents the number of samples used for updating the network. r_i represents the current immediate reward. γ represents the discount factor used to measure future rewards, which controls the degree of importance given to future rewards. The parameters of the actor network are updated using a gradient ascent algorithm based on the output values of the critic network, and hence, the loss function of the actor network is expressed as

$$L^\pi = -\frac{1}{c} \sum_{j \in \{1,2\}} \min_{i=1}^c Q_i^j(s_i, a_i; \beta^{Q^j}) - \alpha \log(a_i). \quad (23)$$

Algorithm 2 Proposed SAC-Based Algorithm

Initialize the network, including the policy network $\pi(s_n; \beta^\pi)$ and two critic networks $Q_1(s_n, a_n; \beta^{Q^1})$, $Q_2(s_n, a_n; \beta^{Q^2})$; The two critic target networks are $\beta^{Q^{1'}} = \beta^{Q^1}$, $\beta^{Q^{2'}} = \beta^{Q^2}$. Initialize experience pool C .

2 **for** each episode **do**
 Initialize the positions of H-UAV and L-UAV k , q_H^0 , q_k^0 and get the initial state s_0 . Randomly initialize the amount of data collected by L-UAV k D_k . Initialize the federated learning model.

4 **for** each communication round n **do**
 The policy network outputs the mean m and standard deviation σ of actions based on the current state s_n .

6 Sample the action a_n from Gaussian distribution $N(m, \sigma)$ and normalize the result to $[-1, 1]$. The normalized actions are then denormalized to their physical values to get v^n , θ^n , I^n , α^n , according to their respective value ranges.

8 **for** each L-UAV k (**in parallel**) **do**
 L-UAV k performs I^n local epochs using its local training data.

10 L-UAV k updates its local model via backpropagation according to equation 3. Translocate to the next position q_k^{n+1} based on the random walk model.

12 Calculate the t_k^{up} , t_k^{down} , t_k^{comp} according to communication and computation model.

end for

14 H-UAV aggregates local models from all L-UAVs according to equation 4 and gets the t_H^{comp} . H-UAV gets its next position q_H^{n+1} using velocity v^n and direction θ^n via equations 13 and 14.

16 Calculate the time t^n by equation 16, then update the state to s^{n+1} . Adopt proposed state normalization (Algorithm 1) to normalize s^n , s^{n+1} .

18 Store this transition (s_n, a_n, r_n, s_{n+1}) into the experience pool.

end for

20 **if** C is large enough **then**
 Randomly sample minibatch transitions c from the experience pool.

22 Update the critic networks, the actor network, the temperature coefficient, and two target critic networks by equation 21, 23, 24, 25, respectively.

end if

24 **end for**

The temperature coefficient is updated based on the difference between the current entropy and target entropy H^{tar} . The loss function is given by

$$L^\alpha = \alpha (-\log(a_i) - H^{\text{tar}}). \quad (24)$$

The two target critic networks are softly updated by

$$\begin{aligned} \beta^{Q^{1'}} &\leftarrow (1 - \tau) \beta^{Q^{1'}} + \tau \beta^{Q^1} \\ \beta^{Q^{2'}} &\leftarrow (1 - \tau) \beta^{Q^{2'}} + \tau \beta^{Q^2}. \end{aligned} \quad (25)$$

To ensure the differentiability of the loss function, the policy is reparameterized using the reparameterization trick, denoted as $a_i = m + \mu\sigma$, where m and σ , respectively, represent the mean and standard deviation output by the actor network and μ is a random variable following a standard normal distribution. According to the problem definition, we define an episode as the entire FL process. Algorithm 2 shows the process of the SAC-based optimization algorithm. The flowchat of the proposed SAC algorithm is shown as Fig. 4.

C. Complexity Analysis

This section analyzes the time complexity of the proposed SAC algorithm, encompassing both training and inference phases. The proposed algorithm comprises an actor network, two critic networks, and two target critic networks, all of which employ fully connected (FC) layers. For an FC layer with β_l input neurons and β_{l+1} output neurons, forward propagation computation per sample is $O(2\beta_l\beta_{l+1})$. The computational complexity of backward propagation and gradient update is twice that of forward propagation, i.e., $O(4\beta_l\beta_{l+1})$ [47].

Let L^a and L^c denote the layer counts for the actor and critic networks, respectively, with the l layer neuron counts represented as β_l^a and β_l^c . These networks compute loss functions via (21) and (23), followed by gradient updates. Thus, for a single sample per update with explicit inclusion of input and output layers, the computational complexity (FLOPs) for actor and critic networks is $O\left(\sum_{l=1}^{L_a} 6(|S|\beta_l + \beta_l^a\beta_{l+1}^a + |A|\beta_{L_a})\right)$ and $O\left(6\sum_{l=1}^{L_c} (|S|\beta_l + \beta_l^c\beta_{l+1}^c + \beta_{L_c})\right)$, respectively. In our framework, episodic updates are performed, i.e., after each episode, the networks undergo N updates with the batch size c . The total number of updated episodes is denoted as E_u . The two target critic networks perform soft updates via (25), with a computational complexity of $O(3\beta_l\beta_{l+1})$. Thus, the total training-phase complexity is

$$\mathcal{C}_{\text{train}} = O\left(6E_uN_c\left(\underbrace{\sum_{l=1}^{L_a} (|S|\beta_l + \beta_l^a\beta_{l+1}^a + |A|\beta_{L_a})}_{\text{actor}} + 2\underbrace{\sum_{l=1}^{L_c} (|S|\beta_l + \frac{3}{2}\beta_l^c\beta_{l+1}^c + \beta_{L_c})}_{\text{critics}}\right)\right). \quad (26)$$

During inference, no network updates and critic evaluations occur, so the complexity is

$$\mathcal{C}_{\text{reference}} = O\left(2\sum_{l=1}^{L_a} (|S|\beta_l + \beta_l^a\beta_{l+1}^a + |A|\beta_{L_a})\right). \quad (27)$$

V. SIMULATION RESULTS AND ANALYSIS

A. Experimental Setup

This section introduces the simulation settings related to the proposed scheme and algorithm. For the FL task, we utilize the ResNet18 neural network for image classification tasks, with the CIFAR-10 dataset consisting of 50 000 training images and

TABLE I
SIMULATION PARAMETERS

Notation	Physical meaning	Value
P_0	blade profile power	80W
P_1	induced power	89W
U_{tip}	tip speed of rotor blade	120m/s
v_0	mean rotor induced velocity	4.03m/s
ζ	fuselage drag ratio	0.6
s	rotor solidity	0.05
ρ	air density	1.225kg/m ³
A	rotor disc area	0.5m ²

10 000 test images. We utilized imbalanced data quantities, randomly sampling (800,1000) images from the training set for each L-UAV using sampling algorithms. Data distribution is characterized by the Dirichlet distribution, which is widely used to depict non-IID data. By default, the parameters of the Dirichlet distribution are set to $\eta = 0.3$. As for FL algorithms, we implement classical FedAvg [12], FedProx [29], and FedDyn [48] as baseline FL frameworks.

We set the range of local epoch as [2, 10]. During the local training phase, the training time of L-UAVs is related to factors such as training samples, model size, batch size, and the type of graphics processing unit (GPU). We set the learning rate and batch size in FL to 0.05 and 32, respectively. We set C_k to be equal to 1×10^6 cycles/sample. The H-UAV performs weighted aggregation of models without engaging in model training, so we set C_H to be 100 cycles/bit.

Regarding the communication simulation parameters, the broadcast bandwidth of H-UAV is denoted as $B = 1$ MHz, and the number of subchannels is set as $M = 20$. The subchannel bandwidth as $B_{\text{sub}} = 50$ kHz and the unit-distance channel power gain as $\rho_0 = 1.42 \times 10^{-4}$. Additionally, we set the noise power to $\sigma^2 = -90$ dB, with the transmission powers of all UAVs set to 1 W. Furthermore, the transmitted data sizes S_1 and S_2 are both 2 Mb. For analog beamforming, the main-lobe gain correction factor is set to $\delta = 0.7$, and the default number of antenna elements is denoted as $A_x = A_y = A_z = 3$. Regarding computational parameters, we set the CPU frequency of H-UAV to 2 GHz and the CPU frequency of L-UAV k to $f_k = 1$ GHz.

We employ the random walk model to describe the randomness of the trajectories of L-UAVs [49], [50]. The number of L-UAVs is set to $K = 5$, with a horizontal range of 1000×400 m and a speed of 10 m/s. The speed range of H-UAV is (5,20)m/s. The flight altitude is $h = 140$ m and $H = 150$ m. As for the flying model of rotary-wing UAVs, the flight parameters are chosen based on practical characteristics, which are referred to [39], as detailed in Table I.

In both SAC and baseline algorithms, we adopt the ac architecture. Both the actor and critic networks consist of two FC layers with 256 and 128 neurons, respectively, followed by a rectified linear unit nonlinear activation. We set the learning rates of the actor and critic networks to 0.0003 and 0.001, respectively, striking a balance between convergence speed and stability. The batch size is set to 256. Additionally, the discount factor is set to $\gamma = 0.99$, and the soft update coefficient is set to $\tau = 0.005$.

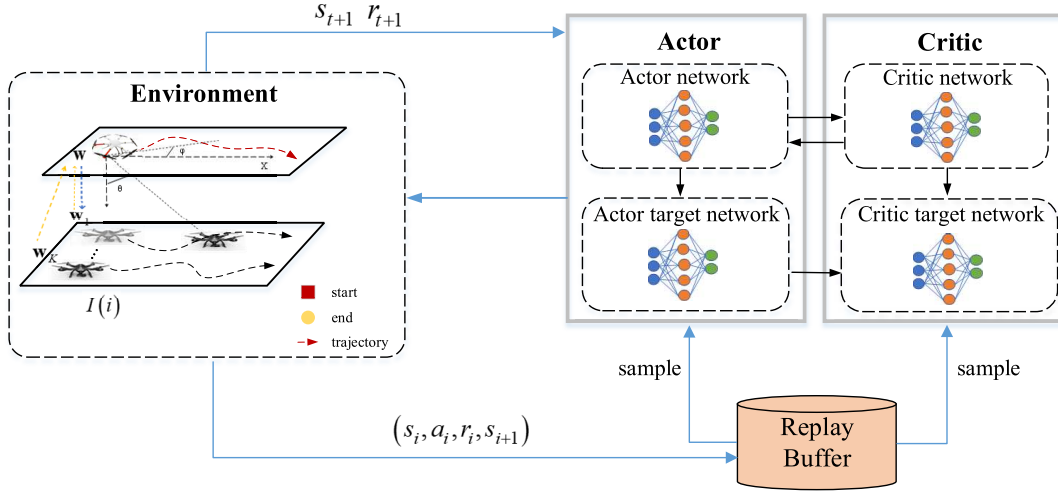


Fig. 4. Flowchart of the proposed SAC algorithm.

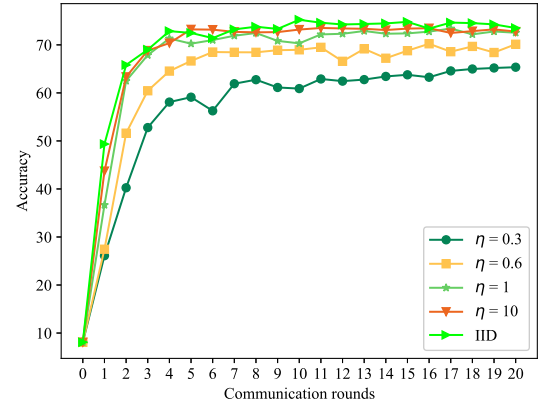
To quantify the respective impacts of these three components (trajectory optimization, beamforming, and FL adjustment) on the optimization objective, we compared our proposed framework against the following three baselines.

- 1) Random trajectory, with only beamforming and FL adjustment.
- 2) Random beamforming, with only trajectory optimization and FL adjustment.
- 3) Random local epoch, with only trajectory optimization and beamforming.

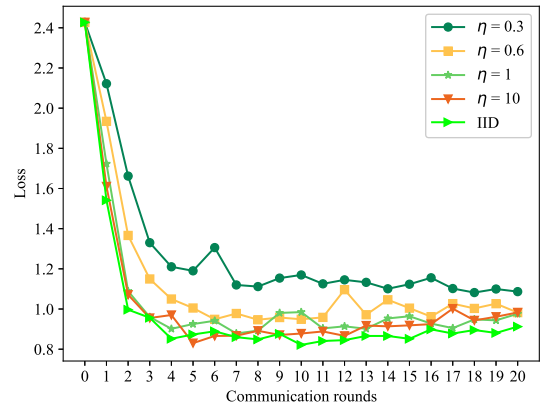
B. Simulation Results of the Proposed Framework

The comparison of model performance under different data distributions with the proposed framework is illustrated in Fig. 5. As shown in Fig. 5(a), the model accuracy improves with increasing communication rounds, though minor fluctuations are observed. The variable η represents the degree of data heterogeneity, where a smaller value indicates a more uneven data distribution, resulting in lower model accuracy. Therefore, as the variable η increases, the model accuracy also correspondingly improves, achieving higher accuracy under non-IID data distributions when $\eta = 10$. Under the IID data distribution, where the data representation from L-UAVs is consistent and stable, the model achieves optimal performance with an accuracy of approximately 74.5%. Moreover, as shown in Fig. 5(b), the model loss gradually decreases, with minor fluctuations caused by the dynamic environment and data heterogeneity. As the degree of data heterogeneity increases, the model loss correspondingly increases.

Fig. 6 illustrates the model accuracy under different numbers of L-UAVs. It can be observed that as the number of L-UAVs increases, the fluctuation of the model accuracy curves decreases, while the model accuracy increases. This is because a greater number of L-UAVs implies a corresponding increase in the data samples used for model training, leading to more diverse data samples and a broader feature space. The data diversity helps the model generalize better across different data distributions, preventing model overfitting and thereby enhancing robustness and improving accuracy.



(a)



(b)

Fig. 5. Model performance comparison of the proposed framework under different data distributions. (a) Model accuracy under different data distributions. (b) Model loss under different data distributions.

Fig. 7 illustrates the relationship between the energy consumption of H-UAV and the number of subchannels under different antenna array designs. As shown in Fig. 7, with the increase in the number of subchannels, the energy consumption gradually decreases. With a greater number of

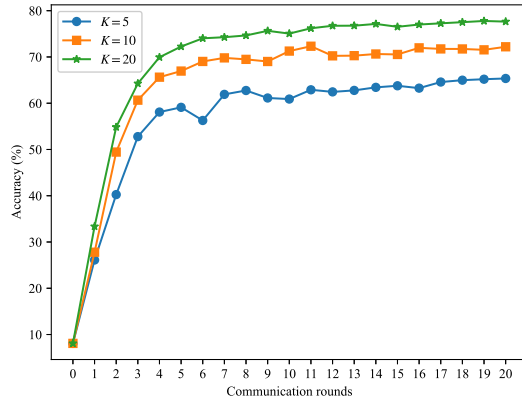


Fig. 6. Model accuracy under the different numbers of L-UAVs.

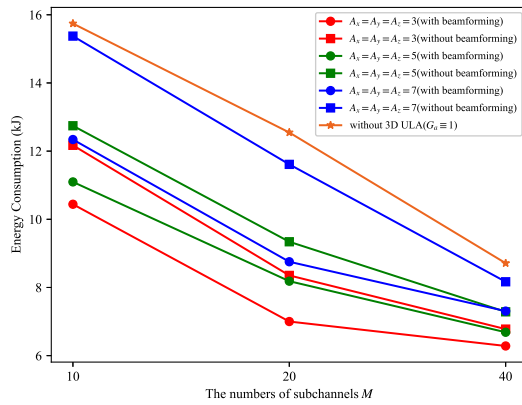


Fig. 7. Energy consumption of H-UAV with different subchannels under different antenna arrays.

subchannels, the uplink channel capacity increases, resulting in shorter transmission time and consequently reduced energy consumption. With the same number of subchannels, the energy consumption is also influenced by the antenna array design. Without 3-D ULA, i.e., $G_a \equiv 1$, it has no spatial diversity gain, so the energy consumption is the highest. Additionally, under the same antenna array design, the energy consumption with 3-D analog beamforming is lower than that without it. This is because it can alter the antenna radiation pattern to improve communication between UAVs. It is worth noting that the more antenna elements there are, the higher the energy consumption. In the setting of FL tasks, due to insufficient data, the H-UAV needs to perform model aggregation after receiving all local model parameters, hence suffering from the transmission time influenced by the L-UAV with the worst channel quality, resulting in the bucket effect. Therefore, although the main-lobe gain increases with the more antenna elements, the radiation width narrows accordingly, and the sidelobe gain usually decreases. Consequently, the transmission time of L-UAVs affected by factors other than the main-lobe gain increases, resulting in higher energy consumption.

Fig. 8 demonstrates the energy consumption comparison between the proposed framework and three baselines under varying numbers of channels M . It can be observed that

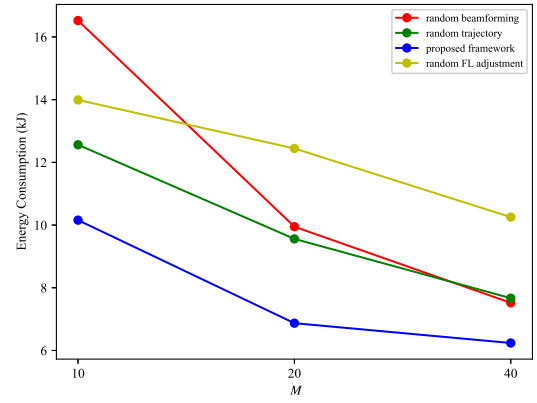


Fig. 8. Training curves for the flight energy consumption.

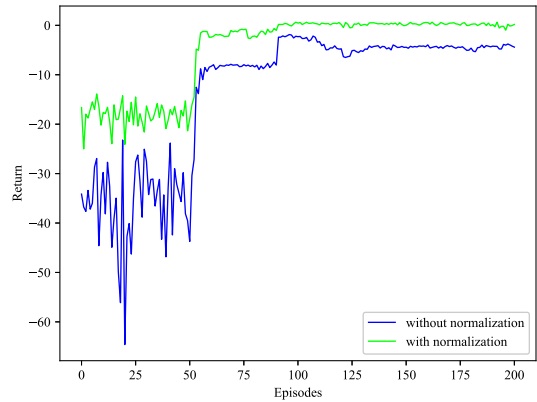
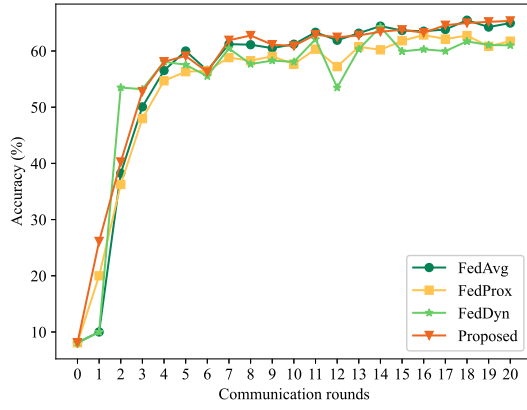


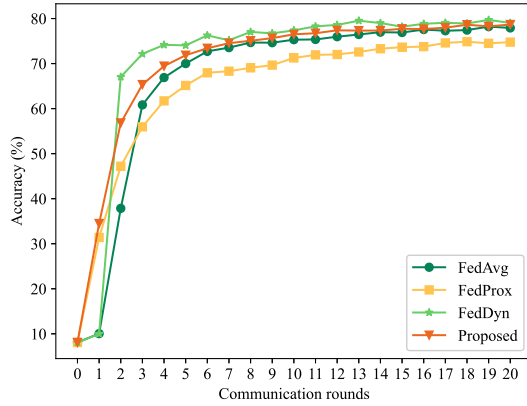
Fig. 9. Training curves for the returns.

the energy consumption of both the proposed framework and the baselines decreases as the number of channels increases, because more available channels reduce communication latency, thereby reducing energy consumption. Furthermore, the proposed framework consistently achieves lower energy consumption than all baselines under different numbers of channels. Notably, the random beamforming baseline exhibits pronounced sensitivity to channel availability. When $M = 10$, it shows the highest energy consumption among the baselines, followed by random local epoch and random trajectory, while with increasing channels, its energy consumption decreases more significantly. When $M = 40$, it consumes less energy than the other two baselines, yet remains higher than the proposed framework.

We compare the training curves of the proposed normalization method with that of the baseline, as shown in Fig. 9. The results show that the unnormalized original state space contains state features with distinct scales, while these scale discrepancies lead to more significant fluctuations, slower convergence, and eventual entrapment in local optima. In contrast, after applying our proposed normalization method, the state features exhibit consistent scales, which prevents certain features from dominating the direction of gradient updates and thereby enhances generalization ability. Consequently, the normalized method achieves faster convergence, higher stability, and superior returns.



(a)



(b)

Fig. 10. Comparison of model accuracy under different FL algorithms. (a) Model accuracy of different FL algorithms when $K = 5$. (b) Model accuracy of different FL algorithms when $K = 20$.

C. Performance Comparison of FL Framework

Fig. 10 presents the comparison between the proposed FL algorithm and the baselines in terms of model accuracy under different numbers of L-UAVs in the proposed scheme. In Fig. 10(a), where $K = 5$ and the proportion of data for training is less than 10% of the full train dataset, it can be observed that the FedDyn algorithm performs the worst with considerable fluctuations. This indicates that the FedDyn algorithm performs poorly with insufficient data. The proposed FL algorithm achieves the overall optimal performance in terms of convergence speed and model accuracy. In Fig. 10(b), as the proportion of data for training increases, the curves under all FL algorithms become smoother and the model accuracy improves by approximately 15% compared to the previous scenarios. It is attributed to the increased data alleviating issues related to fluctuation caused by local model overfitting and low model accuracy due to global model overfitting. In Fig. 10(b), the model accuracy under the proposed FL algorithm achieves 78.67%, slightly lower than 79.7% under the FedDyn algorithm, but with smaller fluctuations. This suggests that the FedDyn algorithm is suitable for scenarios with more L-UAVs and sufficient data. However, the algorithm introduces the penalty risk function with dynamic updates for each FL

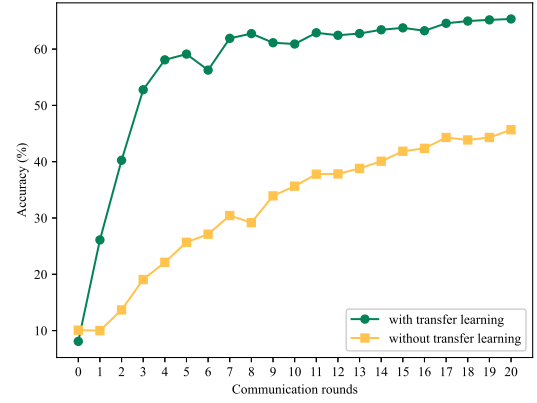


Fig. 11. Model accuracy with or without transfer learning.

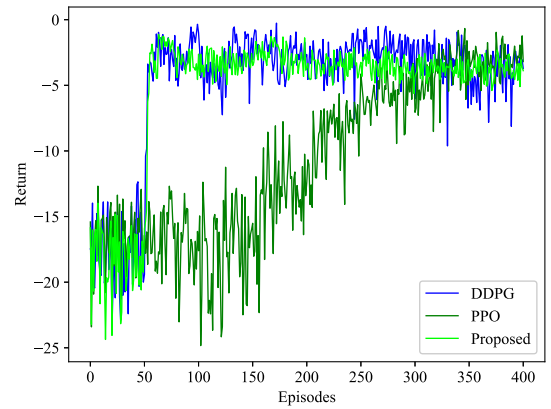


Fig. 12. Training curves for the proposed SAC algorithm and the baseline algorithms.

device, i.e., L-UAVs, which increases computation complexity. Additionally, the proposed algorithm outperforms the FedProx and FedAvg algorithms in terms of model performance, with FedProx exhibiting the worst model performance.

Fig. 11 shows the comparison of model accuracy between the model that employs the transfer learning technique and those that do not. From Fig. 11, it can be observed that despite a slightly lower initial accuracy, the model with transfer learning converges faster, and its accuracy is improved by approximately 20% compared to the model without transfer learning. It is generally challenging to train an efficient model in the absence of sufficient data. Through transfer learning, the abundant source data can be utilized to assist in training the model using sparse target data, which helps the FL model learn more generalized features, mitigates the risk of overfitting when target data are insufficient, and improves model accuracy.

D. Performance Comparison of SAC-Based Algorithm

The curves of return between the proposed SAC-based algorithm and baseline algorithms are illustrated in Fig. 12. Return represents the total reward obtained during the FL task, encompassing both model performance and energy consumption of H-UAV as two metrics. It can be observed that the training

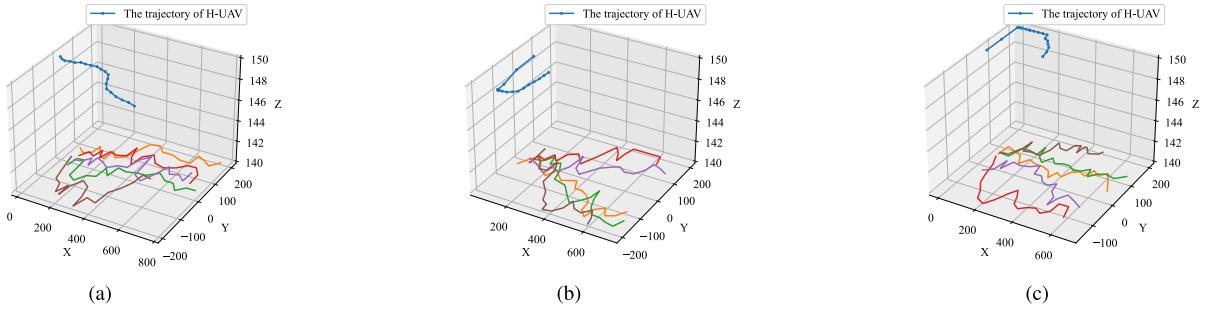


Fig. 13. Illustration of the trajectory for H-UAV between the proposed SAC algorithm and the baseline algorithms. (a) Trajectory under proposed algorithm. (b) Trajectory under DDPG algorithm. (c) Trajectory under PPO algorithm.

curve of the proposed algorithm achieves fast convergence after approximately 60 episodes. This acceleration occurs because the actor network outputs a probability distribution over actions, employing a stochastic policy that enhances exploration by maximizing policy entropy, thereby facilitating faster convergence. Additionally, this curve shows relatively minor fluctuations, which can be attributed to action entropy that promotes action diversity and enhances robustness. In contrast, the DDPG algorithm exhibits poor robustness in highly dynamic environments, resulting in larger fluctuations due to its deterministic actions. Finally, the on-policy property of the PPO algorithm makes it unable to utilize past experience samples, resulting in slower convergence due to insufficient updates. Furthermore, the large variance between the samples for exploration and those for updates leads to pronounced early fluctuations.

Fig. 13 illustrates the trajectories of the H-UAV under different DRL algorithms. Fig. 13(a) presents a trajectory schematic of H-UAV under the proposed SAC algorithm. Despite a slight delay in the flight heading decisions that the H-UAV makes based on information, including the positions of the L-UAVs, the proposed algorithm still plans out a better trajectory, with the H-UAV continuously adjusting its trajectory based on the trajectories of L-UAVs, maintaining its position roughly above the center of the swarm. The trajectory of the H-UAV based on the DDPG algorithm is shown in Fig. 13(b), where the trajectory segment in each time slot is approximately the same, indicating that the H-UAV adopts almost identical flight direction and speed without action diversity. Fig. 13(c) displays the trajectory of the H-UAV under the PPO algorithm. Due to the limitations of the PPO algorithm, it is difficult to adapt to highly dynamic environments in the short term, resulting in local optima and thus poorer trajectories.

Fig. 14 illustrates the model performance of the proposed SAC algorithm compared to baseline algorithms. Based on the proposed algorithm, the model accuracy almost converges to 64% after 6 communication rounds in the FL task. In contrast, the model accuracy achieves 61% after eight communication rounds under the DDPG algorithm, which is approximately 3% points lower than the proposed algorithm. The model performance based on the PPO algorithm almost performs the worst among the overall communication rounds, with model accuracy approximately 4% lower than the proposed algorithm. Correspondingly, the model loss on the test dataset

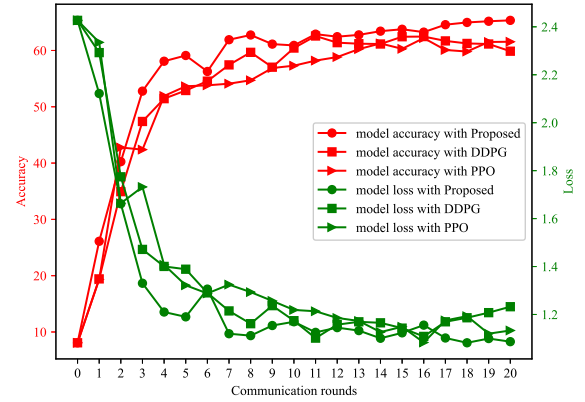


Fig. 14. Comparison of model performance between the proposed SAC algorithm and the baseline algorithms.

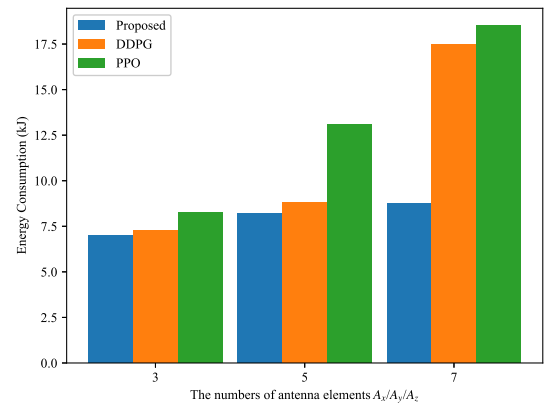


Fig. 15. Comparison of energy consumption between the proposed SAC algorithm and baseline algorithms under the different antenna arrays.

shows lower values, validating the effectiveness of the proposed algorithm.

Fig. 15 presents the comparison of energy consumption for H-UAV under the different DRL algorithms with different antenna arrays. When the number of antenna array elements equals 3, the energy consumption under the three algorithms is approximately with that of the proposed algorithm is the lowest which equals approximately 7 kJ. Furthermore, it can be observed that as the number of antenna array elements increases, the energy consumption under the three algorithms also increases due to the bucket effect. On the other hand,

when the antenna array element equals 7, energy consumption under the two baseline algorithms significantly increases, by more than twofold, while the variation in energy consumption under the proposed algorithm remains relatively small. This also indicates that the proposed algorithm exhibits strong robustness to the environment, as it can mitigate the bucket effect by adjusting the antenna radiation pattern through effective analog beamforming.

VI. CONCLUSION

In this article, we proposed an energy-efficient FL framework in a two-level UAV communication network, including an H-UAV and a group of L-UAVs. To alleviate the problem of insufficient data collected by L-UAVs, we utilized transfer learning techniques to enhance the performance of the FL model. Additionally, we introduced regularization constraints during local training to mitigate the adverse effects of data heterogeneity. Furthermore, we designed local epoch adjustment to balance the training delays and model performance, and enhanced communication efficiency using 3-D analog beamforming. In future work, we aim to develop more robust beamforming techniques that can maintain communication quality despite the high mobility and unpredictable movement of UAVs, as well as secure mechanisms to protect UAV communication against eavesdropping and jamming attacks in sensitive applications.

APPENDIX

CONVERGENCE ANALYSIS OF SAC-BASED FL

We first quantify data heterogeneity by the gradient difference. Define δ_k as

$$\|\nabla f_k(\mathbf{w}) - \nabla F(\mathbf{w})\| \leq \delta_k^{(\tau)} \quad (28)$$

where δ_k represent the data distribution difference between the dataset of the k th L-UAV and the H-UAV at the τ th local iteration. Furthermore, the global-local gradient difference is represented by

$$\delta^{(\tau)} \triangleq \sum_{k \in \mathcal{K}^{(\tau)}} \gamma_k^{(\tau)} \delta_k^{(\tau)} \quad (29)$$

where $\gamma_k \triangleq |\mathcal{D}_k|/|\mathcal{D}|$. The definition indicates that the global-local gradient is time-varying, since the whole dataset of UAV swarm remains the same during training, while the dataset of each L-UAV varies over time due to the mobility of UAVs. Thus, $\delta_k^{(\tau)}$ denotes the data heterogeneity impacted by the mobility of UAVs.

Next, a general convergence bound for FL is introduced based on Theorem 1. Define the following virtual models.

- 1) $u^{(\tau)}$: The virtual global model. It records the weighted sum of local models at each iteration by

$$u^{(\tau)} \triangleq \sum_k \gamma_k \mathbf{w}_k^{(\tau)}. \quad (30)$$

- 2) $\mathbf{v}^{(\tau)}$: The virtual centralized model. It monitors the centralized learning process and regularly synchronizes with the virtual global model. It synchronizes with the

$$\mathbf{v}^{(\tau)} \triangleq \begin{cases} \tilde{\mathbf{v}}^{(\tau)} = \mathbf{v}^{(\tau-1)} - \eta \nabla F(\mathbf{v}^{(\tau-1)}), & \tau_l \tau_e \nmid \tau \\ u^{(\tau)}, & \tau_l \tau_e \mid \tau \end{cases} \quad (31)$$

where τ_l and τ_e are local period and global period, respectively. Here, $a \mid b$ means b is divisible by a , while $a \nmid b$ is not. The upper bound on the central-federal (CF) difference at the n th communication round is denoted by U_n , as

$$\|u^{(n\tau_l\tau_e)} - \tilde{\mathbf{v}}^{(n\tau_l\tau_e)}\| \leq U_n. \quad (32)$$

where U_n represents the difference in the convergence speed between FL and centralized learning. Also, the optimal model parameters are denoted by $\mathbf{w}^*$.

Assumption 1: We assume the following for all L-UAV k .

- 1) $f_k(\mathbf{w})$ is convex.
- 2) $f_k(\mathbf{w})$ is ρ -Lipschitz, i.e., $\|f_k(\mathbf{w}) - f_k(\mathbf{w}')\| \leq \rho \|\mathbf{w} - \mathbf{w}'\|$ for any $\mathbf{w}, \mathbf{w}'$.
- 3) $f_k(\mathbf{w})$ is β -smooth, i.e., $\|\nabla f_k(\mathbf{w}) - \nabla f_k(\mathbf{w}')\| \leq \beta \|\mathbf{w} - \mathbf{w}'\|$ for any $\mathbf{w}, \mathbf{w}'$.

Proposition 1: For any k , if Assumption 1 holds, and for some $\epsilon \geq 0$, we have

- 1) $\lambda \leq 1/\beta$.
- 2) $\lambda\varphi - \rho U_n / \tau_l \tau_e \epsilon^2 > 0$, where $\varphi \triangleq \min_n \|(1 - \beta\lambda/2)/(\tilde{\mathbf{v}}^{(n-1)}\tau_l\tau_e - \mathbf{w}^*)\|^2$ for all n .
- 3) $F(\tilde{\mathbf{v}}^{(n\tau_l\tau_e)}) - F(\mathbf{w}^*) \geq \epsilon$.
- 4) $F(\mathbf{w}^{(n\tau_l\tau_e)}) \geq \epsilon$ for all n .

after $T \triangleq K\tau_l\tau_e$ local updates, the loss function of FL is bounded by

$$F(\mathbf{w}^{(T)}) - F(\mathbf{w}^*) \leq \frac{1}{T\lambda\varphi - \frac{\rho}{\epsilon^2} \sum_{n=1}^N U_n}. \quad (33)$$

We then focus on bounding the CF difference in the mobile scenario. The UAVs have high mobility and may cross different areas.

Theorem 1: In the mobility scenario, the CF difference has an upper bound

$$U_{n,k} = r(\tau_l\tau_e, \lambda, \delta) - \lambda\tau_l \left[\frac{1}{2}\tau_e(\tau_e - 1)\delta - \sum_{j=1}^{\tau_e-1} j\Delta^{[(n-1)\tau_e+j]} \right] \quad (34)$$

where $\Delta^{[j]} = \Delta^{j\tau_l}$, is the loss change for the global model.

Define the mobility factor m_n as

$$\begin{aligned} \gamma_n &= \frac{U_{n,\text{nomob}} - U_{n,\text{mob}}}{\tau_l\tau_e} \\ &= \frac{\lambda}{\tau_e} \left[\sum_{j=1}^{\tau_e-1} j \left(\Delta^{[0]} - \Delta^{[(n-1)\tau_e+j]} \right) - (\delta - \Delta^{[0]}) \frac{H(\tau_l, \tau_e)}{\tau_l} \right] \end{aligned} \quad (35)$$

which characterizes the effect of mobility on the convergence speed at the n th epoch normalized by local updates.

The convergence bound of FL is then expressed by

$$\begin{aligned} F(\mathbf{w}_{\text{mob}}^{(T)}) - F(\mathbf{w}^*) &\leq \frac{1}{T \left(\lambda\varphi - \frac{1}{N} \sum_{n=1}^N \frac{\rho U_{n,\text{mob}}}{\tau_l\tau_e\epsilon^2} \right)} \\ &= \frac{1}{T \left[\lambda\varphi + \frac{\rho}{\epsilon^2} \left(\frac{1}{N} \sum_{n=1}^N \gamma_n - \frac{U}{T} \right) \right]}. \end{aligned} \quad (36)$$

Therefore, the effect of mobility with fixed learning rate λ , T and U is determined by $m \triangleq 1/N \sum_{n=1}^N m_k$. If $m > 0$, mobility accelerates the convergence speed of FL; if $m = 0$, mobility has no apparent influence on the convergence speed; if $m < 0$, mobility degrades the performance of FL.

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